# Sagar Gupta

ProServe (Cloud Consultant) - DevOps/MLOps | Amazon Web Services (AWS) | Hyderabad, India

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## Summary

**Software Engineer (full-stack)** with 2 years at AWS Professional Services as ProServe (Cloud Consultant) - DevOps/MLOps, shipping Python and TypeScript across React, FastAPI, and Node.js. Codify AWS landing zones for enterprise customers in tested Terraform modules. Built and published a Chrome extension (Manifest V3, GraphQL, Cloudflare Workers) on the Chrome Web Store. LeetCode Guardian – 2166 peak rating (top 1.1%), 1300+ problems, 107 contests; Rank 1289 in Google Kick Start Round E '22. Merged 11 upstream PRs into axios, Apache Airflow, lucide, Prefect, Feast, awslabs/mcp, multer, and others. MCA, NIT Warangal (CGPA 8.38).

## Experience

**Amazon Web Services (AWS)** Aug 2024 – Present
*ProServe (Cloud Consultant) - DevOps/MLOps* Hyderabad, India

- **Lead DevOps Consultant, Enterprise customer (Feb 2026 – Present):** Built a Terraform landing zone: Organizations OUs, Control Tower, SCPs, tag policies, account baselines and least-privilege deploy roles.
  - Automated network onboarding and tag-based firewall access with Transit Gateway.
  - Automated account vending with Terraform Cloud and OIDC federation.
- **DevOps Consultant, Global financial services customer (Oct 2024 – Aug 2025):** Wrote tested Terraform modules (terraform test, PyTest) for a multi-account landing zone. Worked on preventative security controls and EventBridge-based cross-account automation.
- **DevOps Consultant, Financial market infrastructure customer (Aug 2025 – Nov 2025):** Refactored legacy Terraform to cloud-native IaC patterns.
- **MLOps Pipeline (SME program, Jun – Aug 2025):** Built the full serving stack for a SageMaker medical image classifier – TensorFlow training jobs, blue-green deployment to a TensorFlow Serving endpoint, a Python Lambda inference API behind API Gateway, a static CloudFront web UI, and Terraform + CodePipeline delivery with drift detection and automatic rollback. Generalized and published by AWS as an official open-source sample (aws-samples, MIT-0).
- **Internal Impact:** Published two more AWS samples, on Organizations governance and an AgentCore agent with governed MCP tools (see Publications). AWS technical field community ambassador (AI/ML, DevOps, CloudOps, Agentic AI, Serverless); Anthropic Claude Ambassador (2026); internal tech talks on Terraform testing and GitOps. Security content review qualification (2026). AWS Jam content tester, DevOps and GenAI (2026).

**Amazon Web Services (AWS)** Mar 2024 – Jul 2024
*Associate Professional Services – DevOps (Internship)* Hyderabad, India

- Shipped an on-premises to AWS ECS migration with blue-green deployment via Terraform, CodePipeline, and CodeDeploy, delivering zero-downtime releases. Instrumented distributed tracing with X-Ray and CloudWatch.

**Ikarus-3D** May 2023 – Jul 2023
*Software Developer Associate (Internship)* Mohali, India

- Containerized a production 3D-asset service on AWS ECS/ECR with a GitHub Actions CI/CD pipeline and integrated SonarQube on EC2 + RDS to enforce code-quality gates on every merge.

## Publications & Recognition

- **Published by AWS:** aws-samples/sample-sagemaker-image-classification-mlops (author, MIT-0) – SageMaker MLOps in 86 Terraform files across 4 stacks and 13 modules: clinical quality and Fairlearn fairness gates, hourly PSI drift that starts retraining, canary deploys with auto-rollback. *2026*
- **Published by AWS:** aws-samples/sample-aws-terraform-org-governance (author, MIT-0) – Terraform for AWS Organizations governance on Control Tower: six policy types as 20 policy documents, delegated admin for 15 services. *2026*
- **Published by AWS:** aws-samples/sample-agentcore-harness-agent-terraform (author, MIT-0) – AWS architecture agent on the Bedrock AgentCore harness, all in Terraform (71 resources): 9 MCP tools behind a default-deny Cedar policy, served to Claude Code, Codex and Kiro; promoted to PROD only after an evaluation passes. *2026*
- **LeetCode Guardian** (top 1.1%) – 2166 peak rating, 1300+ problems, 107 contests. GeeksforGeeks 500+ problems; HackerRank 6-Star Problem Solving.
- Rank 1289 – Google Kick Start Round E '22. *Certificate*

## Industry Certifications (5)

**AWS**: Solutions Architect – Associate | Developer – Associate | Machine Learning Engineer – Associate
**AWS**: AI Practitioner | Cloud Practitioner

Personal projects and opinions are my own and do not represent Amazon or AWS.

## Projects

**Ledger Sync** Jan 2026
*React 19, TypeScript, FastAPI, SQLAlchemy, Neon PostgreSQL, Vercel* GitHub | Live
◇ Self-hosted personal-finance app: imports .xlsx, .xls and .csv bank statements with SHA-256 transaction IDs so re-imports stay idempotent, then serves 26 analytics pages (Indian fiscal-year tax planning, investments, FIRE, multi-currency) and an AI assistant limited to 15 read-only tool calls. FastAPI on Vercel over Neon PostgreSQL, React on GitHub Pages, with 298 test files across both stacks.

**orrery** Oct 2026
*Agent Skill, Claude Code Plugin, Node.js, ELK, SVG* GitHub | Gallery
◇ Agent Skill that turns a YAML spec into animated architecture diagrams with 18,209 official icons (AWS, Azure, Google Cloud, Kubernetes, GitHub, IBM), each cloud pack rebuilt from the vendor's own download and pinned by checksum. ELK auto-layout, a validator whose every finding names the fix, and a review board of 3 reviewer agents; 40 reference diagrams are rendered and checked in CI. Runs in Claude Code, Codex, Cursor, Gemini CLI and Kiro.

**Kinfolk** Jul 2026
*TypeScript, Next.js 16, React 19, Drizzle, Neon Postgres* GitHub | Live
◇ Collaborative family graph modelled as a directed acyclic graph, not a tree: each relative keeps their own record and consent links join them, so nothing is overwritten. Live on Vercel + Neon with a production health check; regression tests run on an in-memory PGlite database.

**GitScope** Mar 2026
*JavaScript, Chrome Extension (Manifest V3), GitHub GraphQL, Cloudflare Workers* GitHub | Chrome Web Store
◇ Published Chrome extension adding GitHub profile insights (9-stat grid, language breakdown, activity heatmap, profile comparison). OAuth runs in a Cloudflare Worker so secrets stay server-side, with 5-minute caching and zero runtime dependencies.

**LeetCode Rating Predictor** Dec 2023
*Python, TensorFlow, NumPy, FastAPI, React, Docker* GitHub | Live
◇ Rating model on 15 engineered features. Found and fixed a train/serve skew on participant count that had tripled backtest error (MAE 38.76 to 12.46 on 57 contests). Moved serving to NumPy-only inference pinned by a golden test, cutting the runtime from ~1.8 GB to ~65 MB; a weekly job refreshes the data and re-runs the backtest.

**Other Notable Projects:** itr-agent (deterministic tax engine, 151 tests), Code Arena (competitive coding arena frontend), Bedrock Multi-Model MCP (26 text models, 156 of 156 benchmark calls), Claude Cost Optimizer (cache-aware Claude cost model), skillcheck (Agent Skills linter, 284 npm downloads a month).

## Open Source Contributions (11 merged upstream)

- **axios** (109K☆) – AxiosError message now built from AggregateError sub-errors, ending blank dual-stack errors (#11059)
- **Apache Airflow** (47K☆) – template_fields on SalesforceBulkOperator, so its arguments render via Jinja at run time (#63109)
- **lucide** (25K☆) – Fixed a TypeError crash when Vue templates pass the kebab-case icon-node prop (#4512)
- **multer** (12K☆) – Files rejected by fileFilter no longer use up maxCount and fail later valid uploads (#1426)
- **terraform-aws-modules/atlantis** – Fixed custom ECS security group egress rules being silently replaced by allow-all (#446)
- **Also merged:** PrefectHQ/prefect, feast-dev/feast, awslabs/agent-plugins (#132, #212), awslabs/mcp, cloudposse/terraform-aws-tfstate-backend; co-author of the aws-ia/terraform-aws-vpc v4.8.0 fix.

## Education

**National Institute of Technology Warangal, Telangana** Sep 2021 – Jul 2024
Master of Computer Applications (MCA) | NIMCET AIR 208 CGPA: 8.38

**Sanskar College of Professional Studies (DAVV University), Indore** May 2018 – Aug 2021
Bachelor of Computer Applications (BCA) CGPA: 7.34

## Technical Skills

**Languages**: Python, TypeScript, JavaScript, C++, Java, Go, SQL, HCL, Bash
**Frontend**: React, Next.js, Preact, Tailwind CSS, Vite, Chrome Extensions (Manifest V3)
**Backend & APIs**: FastAPI, Node.js, Express, GraphQL, REST, WebSockets, gRPC
**Data & Storage**: PostgreSQL, MongoDB, SQLite, Redis, DynamoDB, schema design, query optimization
**Systems & Infra**: Docker, GitHub Actions, AWS (ECS, Lambda, API Gateway, S3, CloudFront, Network Firewall, AWS Backup, Systems Manager Patch Manager, Amazon Bedrock AgentCore), Terraform, Linux
**CS Foundations**: Data Structures & Algorithms, System Design, OOP, OS, DBMS, Computer Networks

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